COMPSCI 308 Design and Analysis of Algorithms

Quiz 03

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2026-02-13

👀 Don't Cheat

During the quiz/exam do *not*

- 🩹 give or receive aid;
- ⏱️ read the questions before the quiz begins;
- 🛑 keep writing when the quiz ends;
- 📱 use a phone, calculator, computer or the internet;
- 🚽 hide textbooks or other aids in the toilet.

⚡ Follow Rules

- 👨‍🎓 Enter your name and **SID** (not your Net ID) in the designated space below.
- 🆎 Support your proofs with both equations and descriptive reasoning.
- ✏️ Use only pen for writing as pencil marks are not clearly visible.
- ❓ Restrict your inquiries to clarifications regarding typos.

Name: SID:

Figure 1: Good Luck

1 Unbounded Feasible Region (1pt)

Give an example of a linear program in two variables x_1 and x_2 for which the feasible region is not bounded, but the optimal objective value is finite.

Answer:

2 Single-Source Shortest Paths as a Linear Program (1pt)

Given the following weighted directed graph G with source vertex s, write a linear program whose solution gives the shortest-path weights d_v from s to every vertex v.

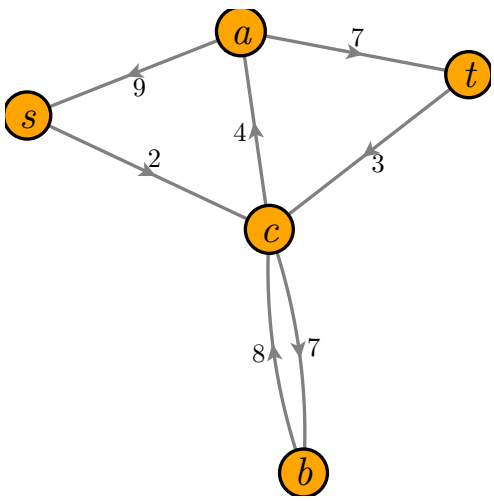

Figure 2: A weighted directed graph with source s.

Write down the LP with an explicit objective function and one constraint per edge, and include the source constraint $d_s = 0$. Use variable names d_s, d_t, d_a, d_b, d_c.

Answer:

3 Maximum Flow as a Linear Program (1pt)

Write out explicitly the linear program corresponding to finding the maximum flow in the following network.

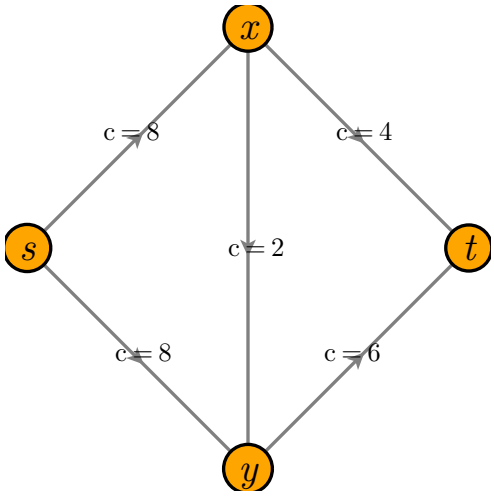

Figure 3: A flow network with edge capacities.

Use a flow variable f_{uv} for each edge (u, v).

Answer:

4 Formulating the Dual (1pt)

Consider the following linear program in standard form:

$$\begin{aligned} \text{maximize} \quad & 8x_1 + 8x_2 + 2x_3 \\ \text{subject to} \quad & 4x_1 + 3x_2 + 4x_3 \leq 13, \\ & 6x_1 + 5x_2 + 3x_3 \leq 26, \\ & 5x_1 + 2x_2 + 1x_3 \leq 17 \\ & x_1,\ x_2,\ x_3 \geq 0. \end{aligned}$$

Formulate its dual.

Answer:

5 Randomized Algorithms (1pt)

Rank

The **rank** of a candidate is their quality relative to all other candidates. A candidate with rank k is the k-th best out of all n candidates. Rank n is the best; rank 1 is the worst.

Consider the following hiring algorithm:

Hire-Detective()
best $= 0$ ▷ A dummy candidate
for $i \leftarrow 1$ to n **do**
  interview candidate i
  if candidate i is better than the current best candidate **then**
    best $= i$
    hire candidate i to be a 🕵

Candidate i	1	2	3	4	5	6	7	8	9	10
Rank	2	9	7	1	4	8	6	10	5	3

How many candidates will be hired to become a 🕵?

Answer:

6 Quicksort (1 pt)

👀 Quicksort and Partition Algorithms

```
QUICKSORT(A, p, r)
if p < r then
    q = PARTITION(A, p, r)
    QUICKSORT(A, p, q − 1)
    QUICKSORT(A, q + 1, r)
```

```
PARTITION(A, p, r)
x = A[r]
i = p − 1
j = p
while j ≤ r − 1 do
    if A[j] ≤ x then
        i = i + 1
        swap A[i] with A[j]
    j = j + 1
swap A[i + 1] with A[r]
return i + 1
```

Trace the execution of PARTITION(A, p, r) on the array below. Show the array after **each iteration** of the while loop and mark the positions of i and j. After the loop, show the **final** array and mark the pivot index q.

💡 The initial state and the state after one iteration are given as examples.

Initial state (pivot $x = 3$)

i	j, p						r
2	5	7	1	4	8	6	3

After iteration 1

i, p	j						r
2	5	7	1	4	8	6	3

After iteration 2

p							r

After iteration 3

p							r

After iteration 4

p							r

After iteration 5

p							r

After iteration 6

p							r

After iteration 7

p							r

Final array

p							r

7 Hash Tables (1pt)

Independent Uniform Hashing

An **ideal** hashing function h maps each key $k \in U$ to a slot in $\{0, 1, \ldots, m-1\}$ so that:

- $h(k)$ is chosen **uniformly** from the m slots,
- choices for different keys are **independent**, and
- once chosen, the value of $h(k)$ stays fixed for that key.

Such a function is called an **independent uniform hash function** (also called a random oracle). Using a hash table under this model is called **independent uniform hashing**.

You use a hash function h to hash n distinct keys into an array T of length m. Assuming independent uniform hashing, let

$$X = |\{\{k_1, k_2\} : k_1 \neq k_2 \text{ and } h(k_1) = h(k_2)\}|.$$

7.1 (0.5 pt)

What is $\mathbb{E}[X]$?

Answer:

7.2 (0.5 pt)

Prove your answer.

Answer:

8 Hash Tables (1pt)

💡 Adapted from Exercise 11.4-1 in *Introduction to Algorithms*.

Consider inserting the keys 65, 37, 7, 1, 96, 94, 75, 83, 64 in the given order into a hash table of length $m = 11$ using open addressing with linear probing $h(k, i) = (k + i) \bmod m$. Give only the final table after all keys are inserted.

Answer:

0	1	2	3	4	5	6	7	8	9	10

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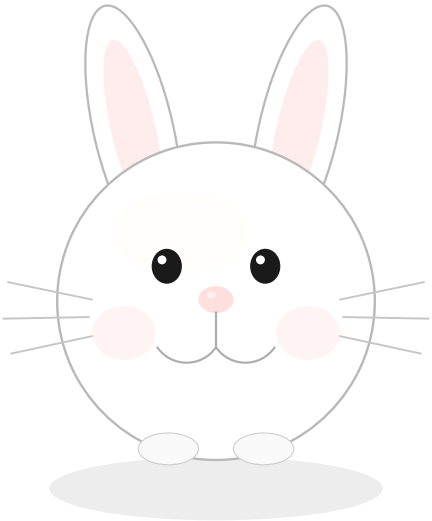
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